Sourav Chakraborty

B.Tech Data Science Student (3rd Year, 5th Semester), Adamas University, Kolkata
souravchakraborty33920@gmail.com | Kolkata, West Bengal | linkedin.com/in/souravchakraborty28 | github.com/SouravCZ

SUMMARY

3rd-year B.Tech Data Science student with hands-on experience building full-stack and data-driven applications, including a production-grade machine learning platform for a national hackathon. Proven ability to lead teams, ship deployed projects, and apply machine learning to real-world problems. Actively building a project portfolio and seeking a Data Science / Analytics internship.

EDUCATION

Adamas University, Kolkata Aug 2024 - May 2028
B.Tech in Computer Science & Engineering (Data Science)
3rd Year, 5th Semester | CGPA: 7.30

PROJECTS

Parivekshan AI - Predictive Land Acquisition Delay Monitoring | *Smart India Hackathon 2026 (PS 26017), Team Lead & Frontend*
github.com/SouravCZ/byte-freaks

- Leading a 5-6 person team ("Byte Freaks") building an AI decision-support platform for District Collectorates to predict land acquisition delays, aligned to RFCTLARR 2013, the National Highways Act 1956, and PM Gati Shakti.
- Trained an XGBoost 4-class risk classifier (Low/Medium/High/Critical) on 29 engineered features, reaching 85.7% accuracy and 0.856 F1, with SHAP explainability surfacing key risk drivers such as compensation gap percentage.
- Built the full-stack frontend (React, Vite, Tailwind CSS, Recharts, MapLibre GL) with a live dashboard, project-level risk gauges, an SLA-based alert queue, and a district-level analytics heat-matrix.
- Live-scores all 66 real West Bengal infrastructure projects against the trained model via a FastAPI ML server proxied through the Node.js/Express and PostGIS backend.

Tech: React, Vite, Tailwind CSS, Node.js, Express, PostgreSQL, PostGIS, Python, XGBoost, SHAP, FastAPI

Student Performance & Attendance Analytics System | *4th Semester Mini Project, Adamas University, Mar 2026 - Apr 2026*
student-analytics-wsy5.vercel.app

- Built and deployed a full-stack analytics platform to track student marks and attendance across subjects and semesters.
- Developed a machine learning engine using Logistic Regression and Random Forest to predict dropout/failure risk with confidence scores for early faculty intervention.
- Implemented an automated alert system that flags at-risk students and generates faculty-facing PDF and CSV reports.

Tech: Flask, React, MySQL, Pandas, Scikit-learn

Customer Churn Prediction | *Personal Project*
github.com/SouravCZ/churn-prediction

- Built an end-to-end classification model to predict customer churn, enabling businesses to proactively identify and retain at-risk customers.
- Delivered as a Flask web application with an interactive SHAP-based explainability dashboard on top of the trained model.
- Completed the full machine learning workflow: EDA, feature engineering, model training, and evaluation using accuracy, precision, recall, and AUC-ROC.

Tech: Python, Flask, Pandas, Scikit-learn, NumPy, SHAP

Credit Risk Scoring | *Personal Machine Learning Learning Project*
github.com/SouravCZ

- Built a structured credit risk pipeline on the South German Credit dataset: EDA, feature engineering, and model comparison between Logistic Regression and XGBoost.
- Added SHAP-based explainability for model predictions and served the trained model through a FastAPI inference endpoint (/predict, /explain).

Tech: Python, Pandas, Scikit-learn, XGBoost, SHAP, FastAPI

Finance Analytics Dashboard | *Personal Project*
github.com/SouravCZ/finance-analytics-system

- Built an interactive web dashboard using Flask and Pandas to analyze spending patterns and visualize financial trends dynamically.
- Implemented dynamic filtering and chart generation to surface actionable insights from transaction data; deployed as a fully functional web app.

Tech: Python, Flask, Pandas, HTML/CSS

SKILLS

Languages: Python, Java, C++, C

Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Logistic Regression, Random Forest

Web & Backend: Flask, FastAPI, React, Node.js, Express, Firebase

Databases & Tools: MySQL, PostgreSQL, PostGIS, MongoDB, Git, GitHub, VS Code

CERTIFICATIONS

- NASSCOM FutureSkills Prime - "Gen AI Tools" Assessment, Gold Category (97%), IT-ITeS Sector Skills Council, MeitY, Aug 2026
- Diploma on Programming Languages - Algorithm, Flowchart, C, C++, Java, Python, HTML, Institute of Intellectual Information

ACHIEVEMENTS & ACTIVITIES

- Winner, Data Wizard 2.0, Maganafest, Adamas University
- Smart India Hackathon 2025 - Qualified internal round as Team Lead
- Clash of Coders 2.0 & 3.0 - Competitive programming participant, Adamas University
- GES - IIT Kharagpur - Participated in 3 events at the Global Entrepreneurship Summit